SHARPE
ENGINEERING
& EQUIPMENT, LLC

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TITLE Westby Creamery
10006 Intake Receiving VFD
10006 VFD-RT00-03
Title Page

DRAWN BY:DOF | SCALE: None
APRV'D BY: | DWG NO: VFD-RT00-03-A01
DATE: 6/24/2026

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TITLE Westby Creamery
10006 Intake Receiving VFD
10006 VFD-RT00-03
Table of Contents

DRAWN BY:DOF | SCALE: None
APRV'D BY: | DWG NO: VFD-RT00-03-A02
DATE: 6/24/2026

ELECTRICAL NOTES:

1. ALL ELECTRICAL WORK SHALL BE IN ACCORDANCE WITH THE LATEST EDITION OF THE NATIONAL ELECTRIC CODE, ALL LOCAL CODES, AND ALL SITE SPECIFIC STANDARDS AND REGULATIONS. IN THE EVENT OF A CONFLICT BETWEEN NATIONAL, LOCAL, AND SITE SPECIFIC CODES, STANDARDS, AND REGULATIONS, THE MORE STRINGENT SHALL APPLY.
2. FIELD LOCATE FINAL POSITIONS OF DEVICES IN COORDINATE WITH SHARPE AND PLANT PERSONNEL.
3. EXPOSED CONDUITS SHALL BE RUN PARALLEL TO OR AT RIGHT ANGLES TO WALLS AND STRUCTURAL MEMBERS.
4. ALL CONDUIT SHALL BE ALUMINUM AND MINIMUM CONDUIT TRADE SIZE SHALL BE 3/4" UNLESS OTHERWISE NOTED.
5. ALL CONDUIT SYSTEMS SHALL HAVE LOW POINT DRAINS INSTALLED.
6. ALL CONDUIT ENTRIES INTO CONTROL CABINETS AND FIELD ENCLOSURES SHALL BE MADE FROM THE BOTTOM. WHERE BOTTOM ENTRY IS NOT FEASIBLE, SIDE CONDUIT ENTRY SHALL BE PERMITTED.
7. ALL WIRING SHALL BE STRANDED COPPER, UNLESS OTHERWISE NOTED.
8. FIELD WIRING TO POWER LOADS SHALL BE THHN (DRY AND DAMP LOCATIONS) OR THWN-2 (WET LOCATIONS), MINIMUM SIZE #14 AWG, UNLESS OTHERWISE NOTED.
9. ALL SINGLE PAIR SHIELDED CABLES SHALL BE #18 AWG, BELDEN #9341.
10. ETHERNET CABLE SHALL BE MINIMUM CAT5.
11. ALL WIRING SHALL BE PROPERLY LABELED. ALL WIRE LABELS SHALL BE BLACK TEXT ON WHITE. WIRE LABEL TEXT SHALL BE IN ACCORDANCE WITH THE ELECTRICAL SCHEMATICS.
12. ANALOG WIRING SHALL NOT BE SPLICED, UNLESS OTHERWISE NOTED.
13. AN EQUIPMENT GROUNDING CONDUCTOR SHALL BE PULLED TO ALL EQUIPMENT. CONDUIT IS NOT ACCEPTABLE AS A GROUND. THE EQUIPMENT GROUNDING CONDUCTOR SHALL BE SIZED PER NEC ARTICLE 250, TABLE 250.122 BASED ON THE RATING OR SETTING OF THE AUTOMATIC OVERCURRENT DEVICE IN THE CIRCUIT AHEAD OF THE EQUIPMENT.
14. ALL METALLIC CONDUITS, CABLE TRAY, SUPPORTS, BOXES AND EQUIPMENT SHALL BE GROUNDED IN ACCORDANCE WITH ALL GOVERNING CODES.
15. ALL FIELD ENCLOSURES, INCLUDING JUNCTION BOXES, SHALL BE NEMA 4X.
16. ALL DISCONNECT SWITCHES SHALL BE MARKED WITH THE EQUIPMENT NUMBER OR NAME OF THE EQUIPMENT INVOLVED.
17. ALL DISCONNECT SWITCHES, CONTACTORS, AND STARTERS SHALL BE 3-POLE, UNLESS OTHERWISE NOTED.
18. DC WIRES AND ANALOG CABLES SHALL BE RUN IN A SEPARATE CONDUIT FROM AC WIRES.
19. MOTOR LOAD CONDUCTORS SHALL INCLUDE AN EQUAL SIZE INSULATED GROUNDING CONDUCTOR FROM THE POWER PANEL, UNLESS OTHERWISE NOTED.

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TITLE Westby Creamery
10006 Intake Receiving VFD
10006 VFD-RT00-03
General Notes

DRAWN BY:DOF | SCALE: None
APRV'D BY: | DWG NO: VFD-RT00-03-A03
DATE: 6/24/2026

CBL	SH	MFG	CAT	REF	DESC1	DESC2	DESC3
CBLN0102	N01	AB	1585J-M8HBJM-3	02			
CBLN0104	N01	AB	1585J-M8HBJM-3	04			
CBLN0106	N01	AB	1585J-M8HBJM-3	06			

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TITLE Westby Creamery
10006 Intake Receiving VFD
10006 VFD-RT00-03
Cable List

DRAWN BY:DOF | SCALE: None
APRV'D BY: | DWG NO: VFD-RT00-03-A04
DATE: 6/24/2026

480VAC @ 100 AMPS ±10%
60 HZ, 3 PHASE
(FLA: 77 AMPS)

DISCB0101
90A

GNDB0100
D167-14
GNDB0103

B01021 B01022 B01023

BK_BN_1AWG BK_OG_1AWG BK_YE_1AWG

TBB0106
PDB370-3

CBB0108
BK_BN_1AWG B01081 to B0200
BK_OG_1AWG B01091 to B0201
BK_YE_14AWG B01101 to B0202

CBB0112
BK_BN_1AWG B01121 to B0300
BK_OG_1AWG B01131 to B0301
BK_YE_14AWG B01141 to B0302

CBB0116
BK_BN_14AWG B01161 to B0400
BK_OG_14AWG B01171 to B0401
BK_YE_14AWG B01181 to B0402

to B0125 B01023
to B0125 B01022
to B0125 B01021

TBB0106
B01021 from B0124
B01022 from B0124
B01023 from B0124

BK_BN_1AWG FUB0132 2A BK_BN_14AWG B01321
BK_12AWG BK_OG_14AWG B01332
BK_YE_1AWG BK_YE_14AWG

PWSB0132
L1 L2 L3
B01342 GND
+ BU_14AWG
-B01341
GN_16AWG
10A 24VDC

CBB0133 10A
BU_14AWG B01331 to D0100
BUWH_14AWG B01341 to D0100
TB1 B01341 BUWH_16AWG to B0209
TB1 B01341 BUWH_16AWG to B0309
TB1 B01341 to B0409

BK_OG_14AWG FUB0137 2A BK_OG_14AWG
BK_YE_14AWG FUB0137 2A BK_YE_14AWG

H4 H2 H3 H1
TB0140 750VA
GN_12AWG B01411
X2 X1 B01372
GND

RD_8AWG FUB0142 10A RD_8AWG B01421 to C0100
WH_8AWG B01411 to C0100

BK_OG_14AWG FUB0145 3A BK_OG_14AWG B01451
BK_YE_14AWG FUB0145 3A BK_YE_14AWG B01461
ACB0145
GN_14AWG
GND
2000BTU

B01023 to B0200
B01021 B01022 B01023
B01022 to B0200
B01021 to B0200

LEGEND:
PANEL WIRING
FIELD WIRING
MULTI-CONDUCTOR CABLE
PNEUMATIC AIR LINE

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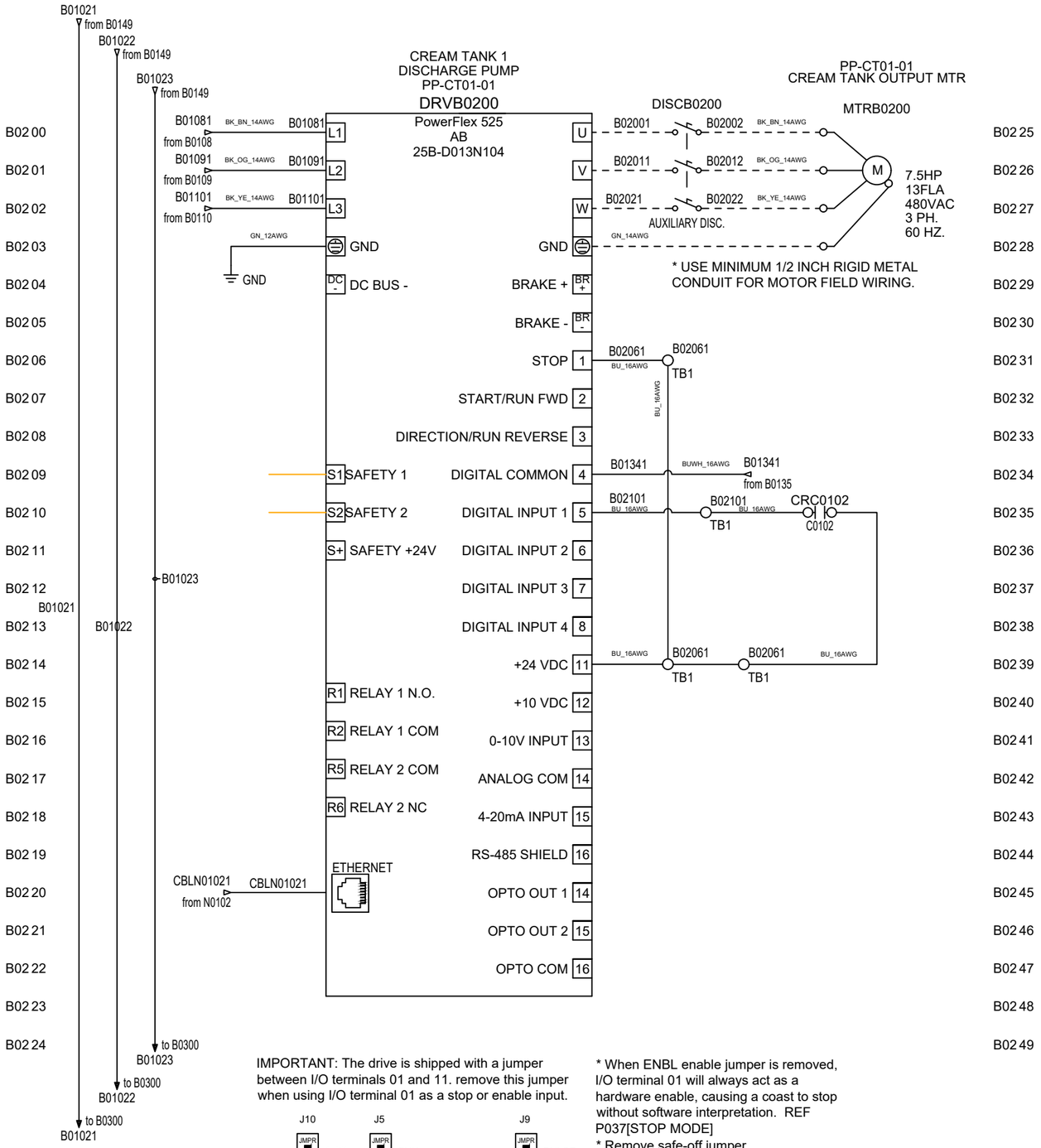
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TITLE Westby Creamery
10006 Intake Receiving VFD
10006 VFD-RT00-03
480VAC Schematics 1

DRAWN BY:DOF
SCALE: None
APRV'D BY:
DWG NO: VFD-RT00-03-B01
DATE: 6/24/2026



IMPORTANT: The drive is shipped with a jumper between I/O terminals 01 and 11. remove this jumper when using I/O terminal 01 as a stop or enable input.

J10 JMPR 0-20V 0-10V

J5 JMPR SNK SRC

J9 JMPR PULSE DIGITAL INPUT

* When ENBL enable jumper is removed, I/O terminal 01 will always act as a hardware enable, causing a coast to stop without software interpretation. REF P037[STOP MODE]
* Remove safe-off jumper

LEGEND:

- PANEL WIRING
- FIELD WIRING
- MULTI-CONDUCTOR CABLE
- PNEUMATIC AIR LINE

CODE	NAME	VALUE	UNITS
P30	LANGUAGE	ENGLISH	
P31	MOTOR NP VOLTS	480	V
P32	MOTOR NP HERTZ	60	HZ
P33	MOTOR OL CURRENT	25	A
P34	MOTOR NP FLA	17	A
P35	MOTOR NP POLES	4	
P36	MOTOR NP RPM	1756	PRM
P37	MOTOR NP POWER	15	HP
P41	ACCEL TIME 1	6	SEC
P42	DECEL TIME 1	0.5	SEC
P46	START SOURCE 1	ETHERNET/IP	
P47	SPEED REFERENCE 1	ETHERNET/IP	
C128	EN ADDRESS DEL	PARAMETERS	
C129	EN IP ADDR CFG 1	192	
C130	EN IP ADDR CFG 2	168	
C131	EN IP ADDR CFG 3	1	
C132	EN IP ADDR CFG 4	60	
C133	EN SUBNET CFG 1	255	
C134	EN SUBNET CFG 2	255	
C135	EN SUBNET CFG 3	255	
C136	EN SUBNET CFG 4	0	

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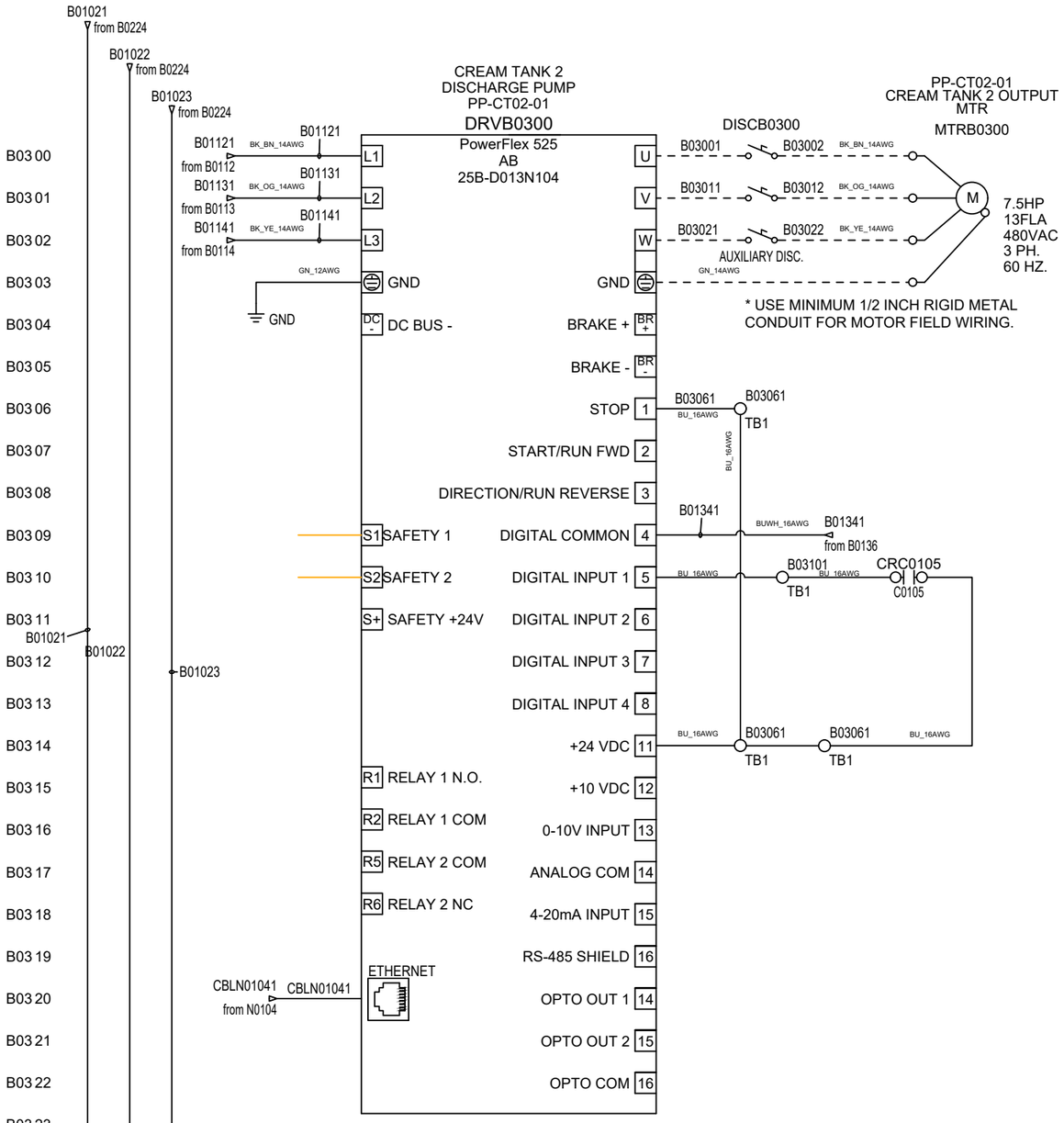
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TITLE Westby Creamery
10006 Intake Receiving VFD
10006 VFD-RT00-03
480VAC Schematics 2

DRAWN BY:DOF | SCALE: None
APRV'D BY: | DWG NO: VFD-RT00-03-B02
DATE: 6/24/2026



CODE	NAME	VALUE	UNITS
P30	LANGUAGE	ENGLISH	
P31	MOTOR NP VOLTS	480	V
P32	MOTOR NP HERTZ	60	HZ
P33	MOTOR OL CURRENT	25	A
P34	MOTOR NP FLA	17	A
P35	MOTOR NP POLES	4	
P36	MOTOR NP RPM	1756	PRM
P37	MOTOR NP POWER	15	HP
P41	ACCEL TIME 1	6	SEC
P42	DECEL TIME 1	0.5	SEC
P46	START SOURCE 1	ETHERNET/IP	
P47	SPEED REFERENCE 1	ETHERNET/IP	
C128	EN ADDRESS DEL	PARAMETERS	
C129	EN IP ADDR CFG 1	192	
C130	EN IP ADDR CFG 2	168	
C131	EN IP ADDR CFG 3	1	
C132	EN IP ADDR CFG 4	61	
C133	EN SUBNET CFG 1	255	
C134	EN SUBNET CFG 2	255	
C135	EN SUBNET CFG 3	255	
C136	EN SUBNET CFG 4	0	

IMPORTANT: The drive is shipped with a jumper between I/O terminals 01 and 11. remove this jumper when using I/O terminal 01 as a stop or enable input.

J10 JMPR 0-20V 0-10V

J5 JMPR SNK SRC

J9 JMPR PULSE DIGITAL INPUT

* When ENBL enable jumper is removed, I/O terminal 01 will always act as a hardware enable, causing a coast to stop without software interpretation. REF P037[STOP MODE]
* Remove safe-off jumper

LEGEND:

PANEL WIRING
FIELD WIRING
MULTI-CONDUCTOR CABLE
PNEUMATIC AIR LINE

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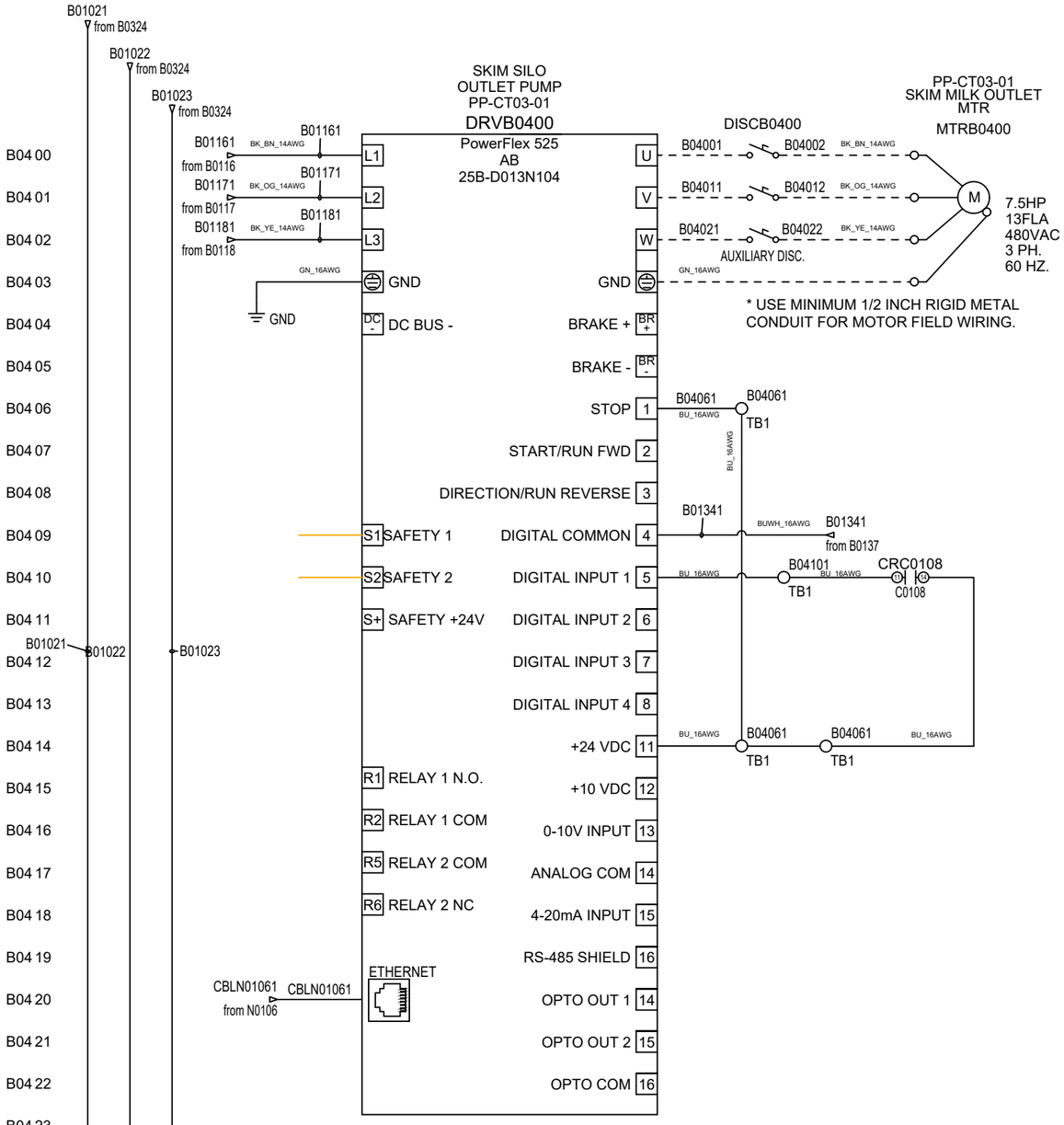
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TITLE Westby Creamery
10006 Intake Receiving VFD
10006 VFD-RT00-03
480VAC Schematics 3

DRAWN BY:DOF | SCALE: None
APRV'D BY: | DWG NO: VFD-RT00-03-B03
DATE: 6/24/2026



CODE	NAME	VALUE	UNITS
P30	LANGUAGE	ENGLISH	
P31	MOTOR NP VOLTS	480VAC	V
P32	MOTOR NP HERTZ	60	HZ
P33	MOTOR OL CURRENT	7	A
P34	MOTOR NP FLA	4	A
P35	MOTOR NP POLES	4	
P36	MOTOR NP RPM	1756	PRM
P37	MOTOR NP POWER	3	HP
P41	ACCEL TIME 1	6	SEC
P42	DECEL TIME 1	0.5	SEC
P46	START SOURCE 1	ETHERNET/IP	
P47	SPEED REFERENCE 1	ETHERNET/IP	
C128	EN ADDRESS DEL	PARAMETERS	
C129	EN IP ADDR CFG 1	192	
C130	EN IP ADDR CFG 2	168	
C131	EN IP ADDR CFG 3	1	
C132	EN IP ADDR CFG 4	62	
C133	EN SUBNET CFG 1	255	
C134	EN SUBNET CFG 2	255	
C135	EN SUBNET CFG 3	255	
C136	EN SUBNET CFG 4	0	

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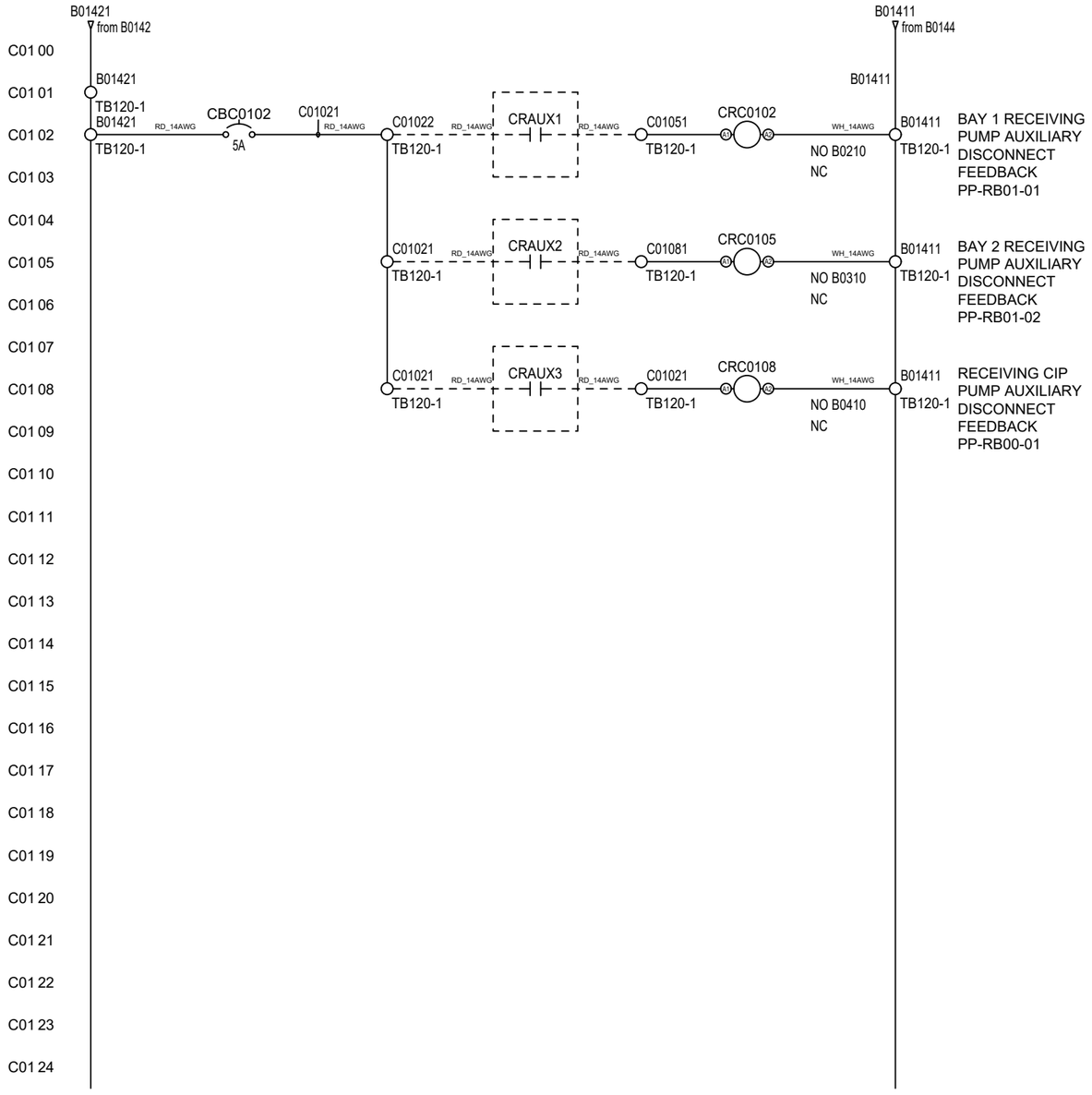
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TITLE Westby Creamery
10006 Intake Receiving VFD
10006 VFD-RT00-03
480VAC Schematics 4

DRAWN BY:DOF | SCALE: None
APRV'D BY: | DWG NO: VFD-RT00-03-B04
DATE: 6/24/2026



LEGEND:

- PANEL WIRING
- FIELD WIRING
- MULTI-CONDUCTOR CABLE
- PNEUMATIC AIR LINE

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TITLE Westby Creamery
10006 Intake Receiving VFD
10006 VFD-RT00-03
120VAC Schematics 1

DRAWN BY:DOF
APRV'D BY:
DATE: 6/24/2026

SCALE: None
DWG NO: VFD-RT00-03-C01

B01331
from B0133

B01331

D01 00 — B01331 — TB24-1

D01 01 — B01331 — TB24-1

D01 02 — BU 14AWG

D01 03

D01 04

D01 05

D01 06

D01 07

D01 08

D01 09

D01 10

D01 11

D01 12

D01 13

D01 14

D01 15

D01 16

D01 17

D01 18

D01 19

D01 20

D01 21

D01 22

D01 23

D01 24

B01341
from B0134

BLWH 14AWG

B01341 — TB24-1

B01341

D01 25

D01 26

D01 27

D01 28

D01 29

D01 30

D01 31

D01 32

D01 33

D01 34

D01 35

D01 36

D01 37

D01 38

D01 39

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D01 43

D01 44

D01 45

D01 46

D01 47

D01 48

D01 49

LEGEND:

- PANEL WIRING
- FIELD WIRING
- MULTI-CONDUCTOR CABLE
- PNEUMATIC AIR LINE

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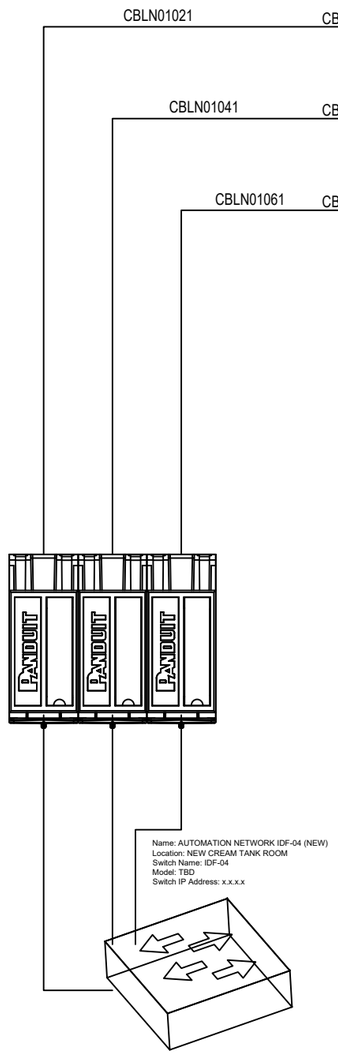
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TITLE Westby Creamery
10006 Intake Receiving VFD
10006 VFD-RT00-03
24VDC Schematics 1

DRAWN BY:DOF | SCALE: None
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DATE: 6/24/2026

N01 00
N01 01
N01 02
N01 03
N01 04
N01 05
N01 06
N01 07
N01 08
N01 09
N01 10
N01 11
N01 12
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N01 44
N01 45
N01 46
N01 47
N01 48
N01 49

LEGEND:

PANEL WIRING
FIELD WIRING
MULTI-CONDUCTOR CABLE
PNEUMATIC AIR LINE

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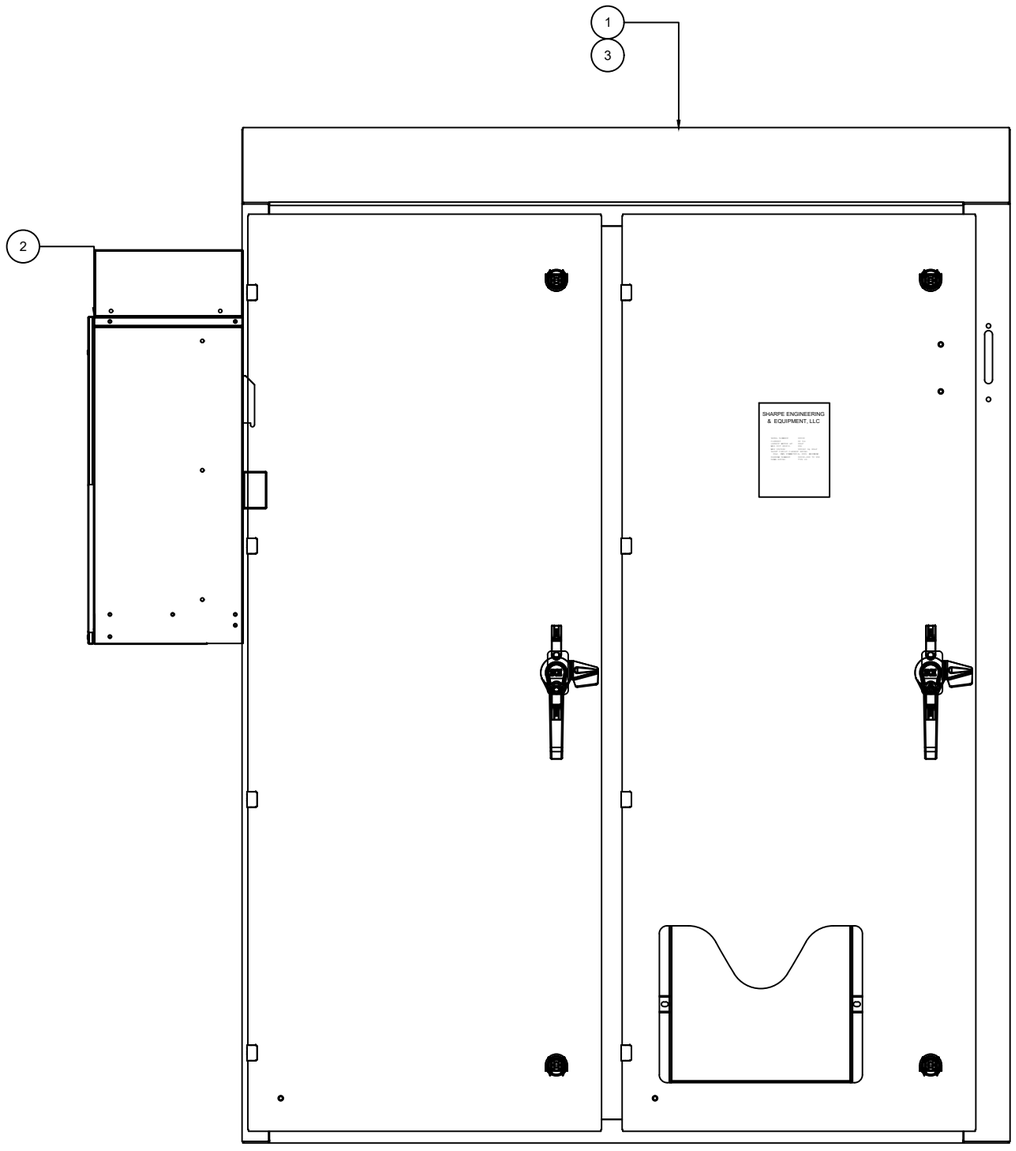
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.XXX = ± .005
.XXXX = ± .0005
ANGLES = ± 1°
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TITLE Westby Creamery
10006 Intake Receiving VFD
10006 VFD-RT00-03
Network Schematics 1

DRAWN BY:DOF
SCALE: None
APRV'D BY:
DWG NO: VFD-RT00-03-N01
DATE: 6/24/2026

BILL OF MATERIAL

ITEM	QTY	MFG	DESCRIPTION	MFG #	SEE#
1	1	SCE	S.S. 2DR XEL Sloping Top Enclosure, 60"x49"x12"	SCE-60XEL4912SSST	PO#
2	1	SHARPE	Panel Name Tag	-	-
3	1	ROXTEC	GLAND, CABLE, 4 PORT, EZ-ENTRY	EZ00000000044	PO#



NOTES:

1) ENCLOSURE LABELED AS FOLLOWING:

SERIAL NUMBER:	VFD-RT00-03
CURRENT:	60A
LARGEST MOTOR HP:	7.5HP
MAX OCP DEVICE:	20 A
VOLTAGE:	480 VAC
PHASE AND FREQUENCY:	3PH 60Hz
SHORT CIRCUIT CURRENT RATING (SCCR)	65kA RMS SYMMETRICAL, 480V MAXIMUM
DIAGRAM NUMBER:	10006 VFD-RT00-03 A01-Z05
NEMA RATING:	TYPE 4X

REV.	DATE	INT.	DETAILS
E			
D			
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INITIAL	06/24/26	DOF	Initial Release

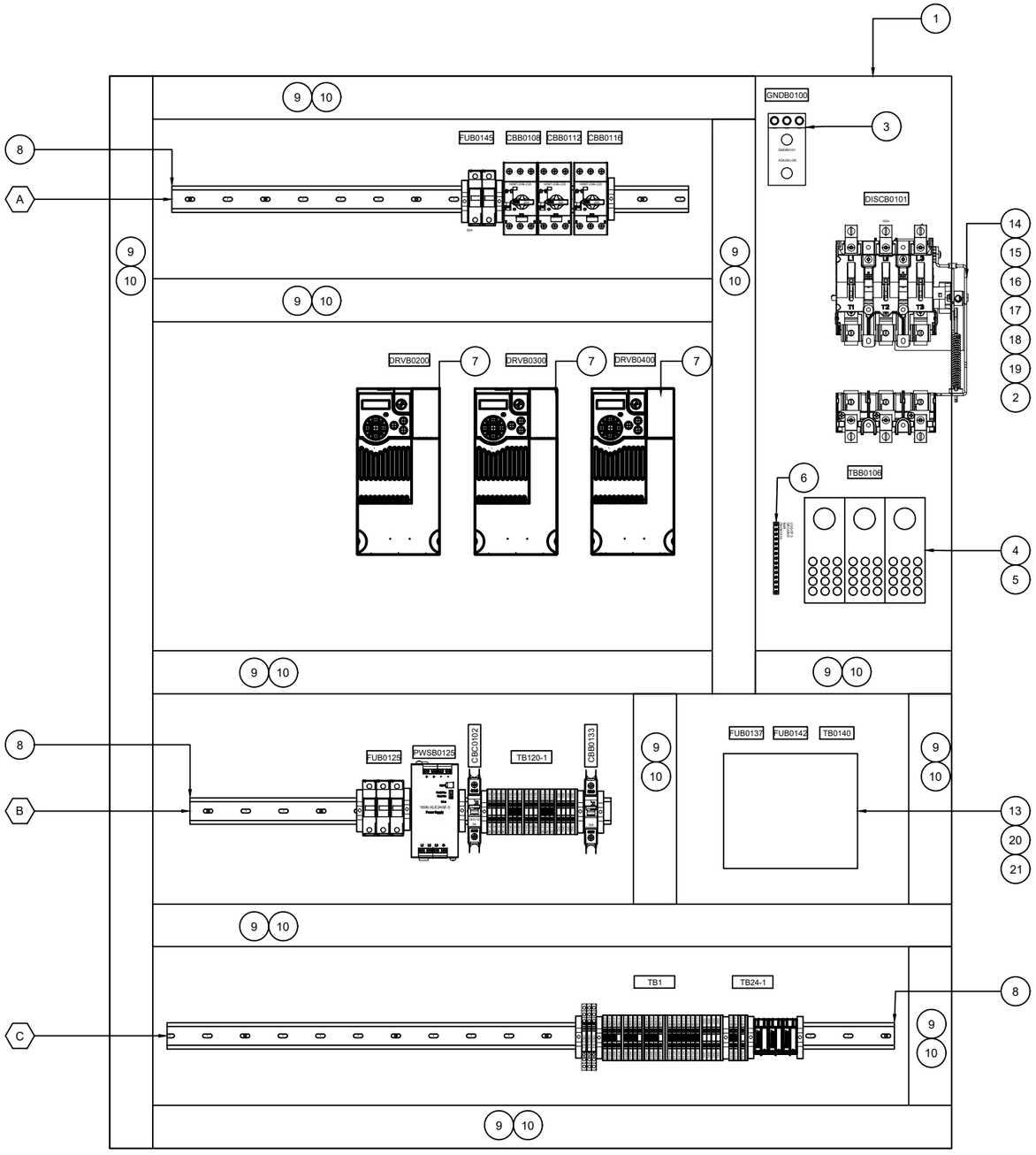
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TITLE Westby Creamery
10006 Intake Receiving VFD
10006 VFD-RT00-03
Panel Layout 1

DRAWN BY:DOF | SCALE: None
APRV'D BY:DOF | DWG NO: VFD-RT00-03-Z01
DATE: 6/24/2026



BILL OF MATERIAL

ITEM	QTY	MFG	DESCRIPTION	MFG #	SEE#
1	1	SCE	Subpanel, Bent, 56"X44"	SCE-60P48	PO#
2	3	BUSSMAN	LPJ fuse, Class J, Time delay, Current-limiting	LPJ-60SP	PO#
3	1	BURNDY	GND LUG	K3A26U-2N	PO#
4	1	AB	POWER DISTRIBUTION BLOCK	1492-PDC-3A200-S0108	PO#
5	1	AB	POWER DISTRIBUTION BLOCK COVER PLATE	1492-PDBC4	PO#
6	1	ILSCO	GND TERMINAL LUGS	D167-14	PO#
7	3	AB	PowerFlex 525 AC Drive, with Embedded EtherNet/IP and Safety, 380...480V AC, 3 Phase, 13 Amps, 7.5 Hp	25B-D013N104	PO#
8	3	AB	DIN RAIL	199-DR1	PO#
9	24'	PANDUIT	2"X3" WIREWAY	G2X3WH6	PO#
10	24'	PANDUIT	PANDUIT 2" COVER WHITE	C2WH6	PO#
11	2	AB	DIN RAIL	199-DR1	PO#
12	1	ILSCO	GND TERMINAL LUGS	D167-14	PO#
13	1	SOLA	750VA 480VAC TO 120VAC TRANSFORMER	E750WB	PO#
14	1	AB	1494U Disconnect Switch Rod or Cable Operated, 60A	1494U-D60	PO#
15	1	AB	1494U Fuse Block, 60A	1494U-F60	PO#
16	1	AB	Threaded rod for 30 to 200A	1494U-R1	PO#
17	1	AB	Stainless Metal handle for 30A to 200A	1494U-HS1	PO#
18	1	AB	1494U Protective Covers, Protective cover, 30A - 60A	1494U-PC1	PO#
19	1	AB	1494U 60A Class H/J 600V Fuse Clip	1494U-FC60J	PO#
20	2	Littlefuse	Littelfuse POWR-GARD CCMR002.TXP Time Delay Fuse, 2 A, 600 V AC	CCMR002.TXP	PO#
21	1	Littlefuse	Littelfuse POWR-GARD CCMR010.TXP Time Delay Fuse, 10 A, 600 V AC/250 V DC,	CCMR010.TXP	PO#

ASSEMBLY REF.	
ASSEMBLY INDEX	REFERENCE
A	Z03
B	Z04
C	Z05

REV.	DATE	INT.	DETAILS
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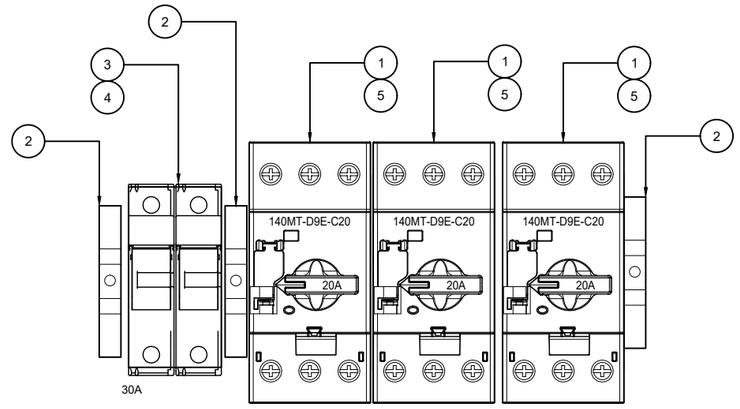
TITLE Westby Creamery
10006 Intake Receiving VFD
10006 VFD-RT00-03
Panel Layout 2

DRAWN BY:DOF | SCALE: None
APRV'D BY: | DWG NO: VFD-RT00-03-Z02
DATE: 6/24/2026

BILL OF MATERIAL

ITEM	QTY	MFG	DESCRIPTION	MFG #	SEE#
1	3	AB	MPCB, Standard Magnetic Trip (Fixed at 14 x Ie), for C36..C40 (12 x Ie), 14.5 - 20 A, High Performance, Frame Size D	140MT-D9E-C20	PO#
2	2	AB	END ANCHOR, DIN RAIL	1492-EAJ35	PO#
3	1	Littlefuse	POWR-SAFE LPSC002ID Dead Front Indicating Fuse Holder With Blown Fuse Indicator, 600 V AC/DC, 30 A, Class: CC, 14 to 8 AWG Wire, 2 Poles	LPSC002ID	PO#
4	2	Littlefuse	Littelfuse POWR-GARD CCMR003.TXP Time Delay Fuse, 3 A, 600 V AC	CCMR003.TXP	PO#
5	3	AB	UL Type E Spacing Adapter, For 140MT, Mtr Protection Ckt-Br C, D Frame	140MT-C-TE	PO#

FUB0145 CBB0108 CBB0112 CBB0116



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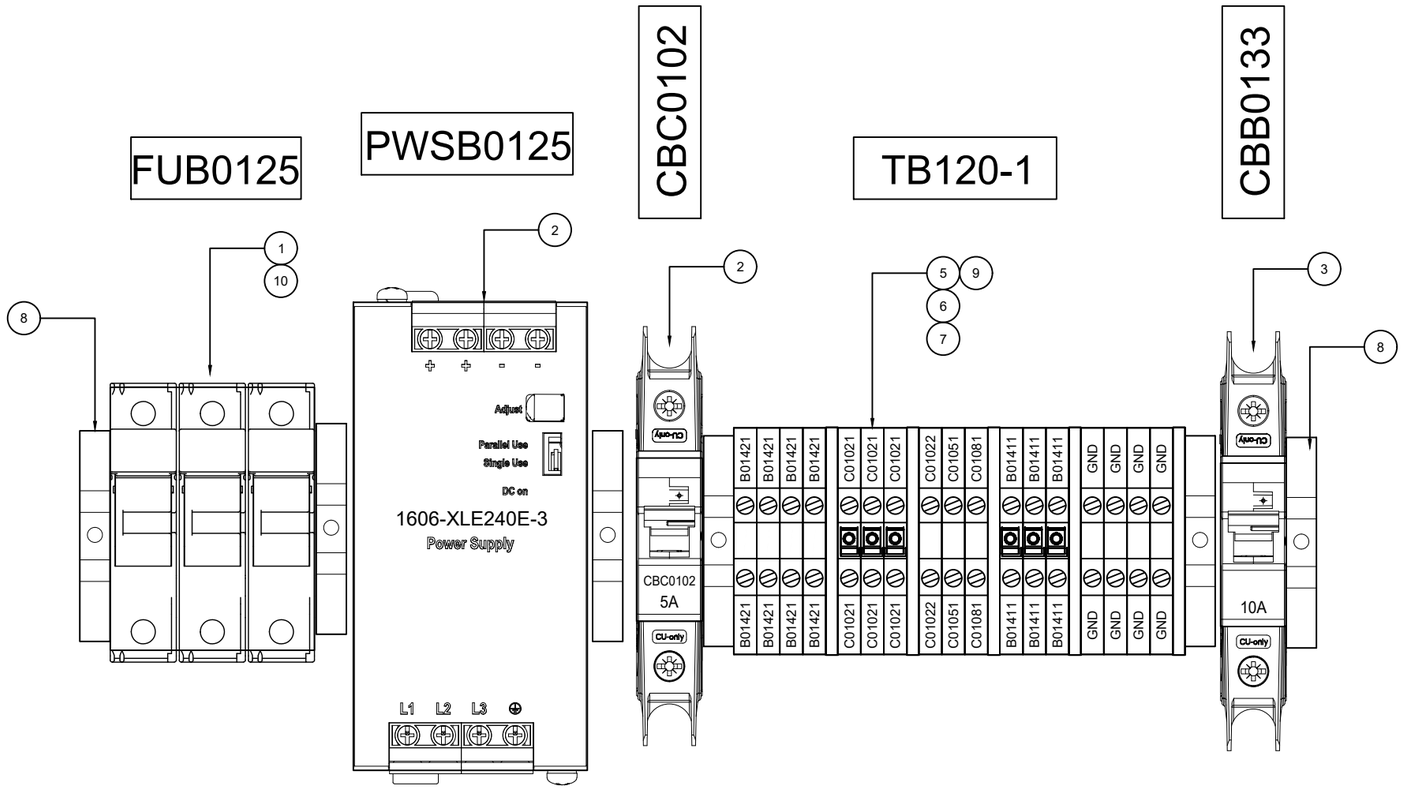
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TITLE Westby Creamery
10006 Intake Receiving VFD
10006 VFD-RT00-03
Panel Layout 3

DRAWN BY:DOF | SCALE: None
APRV'D BY: | DWG NO: VFD-RT00-03-Z03
DATE: 6/24/2026

BILL OF MATERIAL

ITEM	QTY	MFG	DESCRIPTION	MFG #	SEE#
1	1	Littlefuse	POWR-SAFE LPSC002ID Dead Front Indicating Fuse Holder With Blown Fuse Indicator, 600 V AC/DC, 30 A, Class: CC, 14 to 8 AWG Wire, 3 Poles	LPSC003ID	PO#
2	1	AB	24-28VDC, 240W, 3-PHASE 380-480VAC INPUT VOLTAGE	1606-XLE240E-3	PO#
3	2	AB	1489-M Thermal-magnetic Circuit Breakers, Standard configuration, AC, 1 Pole Configuration, Trip Curve C, UL/CSA Max. Voltage: 277V AC, 48V DC ; IEC/EN Max. Voltage: 230V AC,10A	1489-M1C100	PO#
4	1	AB	1489-M Thermal-magnetic Circuit Breakers, Standard configuration, AC, 1 Pole Configuration, Trip Curve C, UL/CSA Max. Voltage: 277V AC, 48V DC ; IEC/EN Max. Voltage: 230V AC,5A	1489-M1C050	PO#
5	15	AB	1492-J IEC Terminal Block, One-Circuit Feed-Through Block	1492-J4	PO#
6	5	AB	End Barrier , Grey (Standard)	1492-EBJ3	PO#
7	4	AB	1492-J IEC Terminal Block, One-Circuit Feed-Through Ground Block	1492-JG4	PO#
8	6	AB	END ANCHOR, DIN RAIL	1492-EAJ35	PO#
9	3	AB	SCREW CENTER JUMPER - 10 POLE FOR 1492-J4	1492-CJJ6-10	PO#
10	3	Littlefuse	Littelfuse POWR-GARD CCMR002.TXP Time Delay Fuse, 2 A, 600 V AC	CCMR002.TXP	PO#



REV.	DATE	INT.	DETAILS
E			
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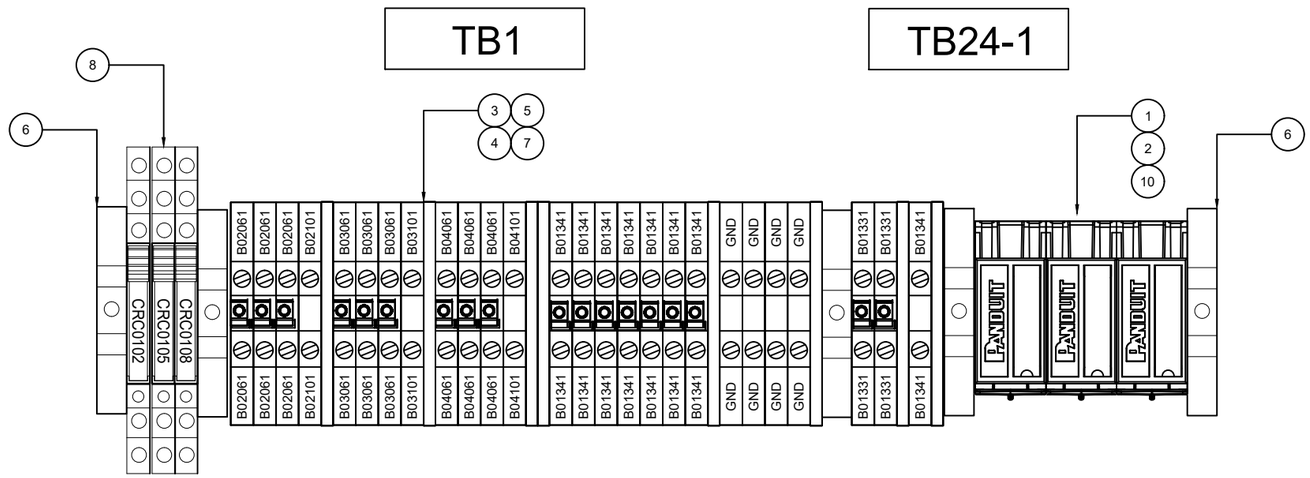
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TITLE Westby Creamery
10006 Intake Receiving VFD
10006 VFD-RT00-03
Panel Layout 4

DRAWN BY:DOF | SCALE: None
APRV'D BY: | DWG NO: VFD-RT00-03-Z04
DATE: 6/24/2026

BILL OF MATERIAL

ITEM	QTY	MFG	DESCRIPTION	MFG #	SEE#
1	3	AB	1585 Ethernet Cables Cat5e, 100BASE-TX, 100 Mbit/s, 8 Conductors, RJ45, Straight Male, Standard, RJ45, 600V	1585J-M8HBJM-3	PO#
2	3	Panduit	Category 6 RJ45 UTP Mini-Com® Jack, International White.	CJ688TGIW	PO#
3	22	AB	1492-J IEC Terminal Block, One-Circuit Feed-Through Block	1492-J4	PO#
4	7	AB	End Barrier , Grey (Standard)	1492-EBJ3	PO#
5	4	AB	1492-J IEC Terminal Block, One-Circuit Feed-Through Ground Block	1492-JG4	PO#
6	5	AB	END ANCHOR, DIN RAIL	1492-EAJ35	PO#
7	2	AB	SCREW CENTER JUMPER - 10 POLE FOR 1492-J4	1492-CJJ6-10	PO#
8	3	AB	A-B Rockwell 700-HLT1L1 GP Terminal Block Relay 120VAC Coil	700-HLT1L1	PO#
9	3	Panduit	Panduit IndustrialNet CADIN1IG Adapter, 1 Port, DIN Rail Mount	CADIN1IG	PO#



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TITLE Westby Creamery
10006 Intake Receiving VFD
10006 VFD-RT00-03
Panel Layout 5

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APRV'D BY: | DWG NO: VFD-RT00-03-Z05
DATE: 6/24/2026